## Phase 3A | Engineering definition

PULSIPHER FARMS / MOAPA VALLEY CORN MAZE

# Christmas Passenger Train Technical Development Package

PF-TRAIN-S3A-R0 | 17 October 2026

**Issue purpose:** component evaluation, engineering coordination, prototype planning and supplier review. **Not for fabrication, construction, procurement release or passenger operation.**

| This release develops | This release leaves held |
|---|---|
| Actual loads, component evidence, frame load paths and functional system architecture. | Final structural sections, weld sizes, wheel profiles, component selection and approved safety circuits. |
| Coach, coupling, stair, brake, electrical and track/station interface drawings. | Surveyed route and North Pole station position; site-specific civil and platform construction details. |
| Fabrication/inspection sequence, supplier response requirements and editable calculation/source records. | Authority classification, signed engineering, supplier commitments and commissioning acceptance. |

**Owner direction:** Track design target is 0% longitudinal grade; finished elevations and engineer-set tolerances must be verified independently of drainage. North Pole location will be supplied later. Passengers will disembark there. Vehicle development therefore includes safe boarding and accessible boarding at both stops. Ride experience, trip scheduling and business projections are outside this technical issue.

The Stage 2 coach geometry is retained as the working arrangement. The rotating 3D models remain Stage 2 space models; new Phase 3 functional diagrams do not convert them into manufactured assemblies. Original H/I/J and Stage 1/2 records are preserved.

# 01 | Recommended development direction

| Decision | Phase 3A recommendation and evidence still needed |
|---|---|
| Frame and bogies | Develop an engineered steel underframe with purchased engineered bogies as the lead comparison; evaluate lighter body systems. Do not preserve the old 3 in floor/frame slab as a structural depth. Close load paths, fatigue, torsion, body mounts and actual component masses. |
| Propulsion | Evaluate complete engineered locomotives against the entire consist. For custom work, compare diesel mechanical, hydraulic and electric as integrated systems. Separate hotel power from tractive demand and driven mass. |
| Brakes | Retain hydraulic service actuation. Develop per-car stored-energy emergency/parking with supervised continuity as one architecture, and compare automatic railway air braking. No architecture has yet earned a fail-safe designation. |
| Connections | Develop a rated rail coupling or engineered articulated drawbar with positive retention and a frame load path. An ordinary road-trailer connection is not a qualified substitute. |
| Accessible boarding | Develop level/near-level bridge boarding at both stops. Evaluate an onboard lift/rescue option separately; the documented large-lift candidate conflicts with the current 42 in side doorway. |
| Track and site | Continue route-independent vehicle and track interface engineering now. Freeze site-specific alignment, drainage and stations after survey and actual equipment limits are available. |

These are preliminary development choices. Supplier dimensions, ratings, tolerances, lead times and project compatibility must be confirmed in writing before selection. A published product is not a quote or proof of stock.

# 02 | Findings that change the next drawings

| Finding | Required response |
|---|---|
| Thin floor/frame envelope | The old display slab is only 3 in deep over a 216 in bogie-center span. Member arrangement/depth, body participation, equipment clearances and fatigue must be solved before the 28 in floor height is fixed. |
| Uneven passenger loading | Regular stress gross mass is 11,825 kg. Gross/4 gives 2,956 kg per axle only with equal sharing. A longitudinal crowd-shift screen raises one bogie reaction; qualify bogies and axles from actual worst load cases, not the average. |
| Level-track traction | At 65 t and 0% grade, assumed 1% starting resistance requires 6.37 kN before acceleration/curves. Adding 0.10 m/s2 acceleration raises this to 12.87 kN. At assumed adhesion 0.15, the latter requires about 8.75 t driven mass. Actual curves, resistance and driven weight still govern selection. |
| Braking convention | This issue fixes specific brake force B/M, then subtracts downhill gravity to get net deceleration. Stage 2 instead used a chosen net deceleration target. Do not compare the stopping distances without reconciling that definition and the delay inputs. |
| Lift doorway conflict | The documented BraunAbility large-platform candidate needs more than the current 42 in clear opening. A 48 in alternate opening is an unintegrated study, not a revision to the retained 3D CAD. |
| Station length | The four coach envelopes plus three study gaps occupy 140 ft 10 in. This excludes the locomotive, its connection and end margins, and is not a released platform or full-train berth length. |

Calculations are transparent screens using stated assumptions. They identify where more engineering is required; they do not establish operating limits or permit the purchase of safety-critical components.

# 03 | Shared basis and calculation control

| Item | Working value | Status |
|---|---|---|
| Train | 1 locomotive + 3 regular + 1 accessible coach | User configuration / study layout |
| Gauge | 36 in (914.4 mm) | Evaluation basis; wheel/rail system not selected |
| Regular / accessible occupancy | 32 guests / 11 fixed + 2 wheelchair places | Layout study; occupied mockup and approval pending |
| People / devices | 109 guests + 4 total crew; 100 / 145 kg per person; 300 kg devices | Nominal / stress inputs, not passenger weight limits |
| Car tare allocations | Regular 7,040 kg; accessible 7,535 kg | Mass budgets, not measured takeoff |
| Train mass | 54,755 kg nominal; 59,840 kg stress; 55/60/65 t sensitivities | Calculated from allocations |
| Cabin / exterior | 306 x 104 in clear; 312 x 110 in outside | Stage 2 dimensions, subject to body build-up |
| Coach / bogie spacing | 400 in envelope; 216 in bogie centers | Study geometry |
| Floor / platforms | 28 in above rail; 44 in platform at each end | Provisional packaging |
| Longitudinal grade | 0% level-track target | Owner direction; survey and engineer-set acceptance tolerances pending |
| Speed / curves | 5 and 7.5 mph; 30/50/75 m radius | Sensitivity only, not permissible limits |

Crew mass is conservatively allocated one per coach for local screens, while the four-person total includes the driver. Final crew positions, fueled locomotive mass, luggage, devices, equipment masses and centers of gravity must replace these assumptions. Do not count occupants or growth allowance twice.

The 65 t performance sensitivity is not the structural mass rating of the retained coach. The brake analysis distributes that upper mass into approximately 13.171 t per regular coach. Adopting that mass would require revised frame, bogie, axle, brake and body load cases; the 11.825 t regular stress case alone would no longer cover it.

Formulas and case inputs are preserved in the calculation source files. Each discipline distinguishes verified publications, calculated results, analyst assumptions, development choices and held engineering decisions. The source register records the access date and limitation of each publication.

# 04 | Frames, bogies, wheels and stability

PF-TRAIN-S3A-R0 • 17 October 2026 • Preliminary; not for fabrication or construction.

## 1. Recommended architecture and dimensional conflict

Retain two two-axle bogies per coach as the development arrangement. Prefer an engineered, supplier-supported running-gear package with wheelsets, bearings, suspension, brakes and body attachment interfaces specified together. Farm fabrication of unproven wheelsets or bogies is not the starting recommendation. Matching 36-inch gauge establishes only one interface.

Use a welded steel underframe with two continuous longitudinal load paths, transverse bolsters at x=45 and 261 inches, end crossmembers, and framed coupling attachment boxes connected into both longitudinal members. Member sections, spacing, steel grade, plate thickness, fasteners and welds are **held**, not selected. Crossmembers carry the floor, seat anchors and outboard wall supports. End-platform, stair and lift reactions enter reinforced frame nodes. Cladding and decorative trim carry no credited primary loads. A stressed-skin body could reduce underframe mass, but only after analysis includes door/window openings, fatigue, joint stiffness and corrosion; do not quietly credit it in the present chassis calculation.

The 400-inch frame extends from x=-47 to 353; the 312-inch cabin from -3 to 309. Each cabin end overhangs its adjacent bogie centre by 48 inches, and each frame tip by 92 inches. The 44-inch platforms therefore create substantial cantilevers. Coupler loads act through the end-frame structure rather than floor sheet or platform trim.

**The previous 3-inch frame-depth graphic is unengineered.** A 216-inch bogie spacing gives span/depth=72 before accounting for openings and connection details. This ratio is a warning, not proof of failure. Expect to redesign the depth/load path; preserve the 28-inch floor height only if the complete wheel, suspension travel, bolster, frame and floor stack fits. Raising the floor changes stairs, accessible boarding, CG and roof clearances. The CAD envelope must remain adjustable until that stack is resolved.

## 2. Static load ledger and cantilever coefficients

Mass allocations come from the shared design basis, not measured vehicles. A local crew allowance of one person per coach is retained for structural screening; it is not an operating staffing decision. Accessible totals include two wheelchair occupants within the 13 guests, one crew member, and 300 kg total mobility devices.

| Coach / person mass | Gross kg | Weight kN | Balanced bogie kg-equivalent | Balanced axle kg-equivalent |
|---|---|---|---|---|
| Regular / 100 kg | 10,340 | 101.40 | 5,170 | 2,585 |
| Regular / 145 kg | 11,825 | 115.96 | 5,912.5 | 2,956.25 |
| Accessible / 100 kg | 9,235 | 90.56 | 4,617.5 | 2,308.75 |
| Accessible / 145 kg | 9,865 | 96.74 | 4,932.5 | 2,466.25 |

These are equilibrium averages with symmetric tare and payload, **not ratings**. No dynamic factor has been assigned. Equipment, roof HVAC, lift, wheelchair bays, tanks and end-platform loads require individual coordinates. Bogie self-mass is included in the tare allocations: subtract actual bogie mass before distributing body loads onto the frame; add it back when checking rail loads.

For items of mass m at x inches, total front-bogie reaction is $\Sigma m(261-x)/216$ kg-equivalent; rear reaction is total mass minus front. Multiply kg-equivalent by 9.80665 for newtons. As an adverse disembarking sensitivity, moving the regular coach's 33 people to a front-half centroid x=76.5 gives front/rear reactions of

**6,338.75/4,001.25 kg** nominal, or **7,607.19/4,217.81 kg** at 145 kg/person. This is redistribution of the same passenger load, not added occupancy. It is not a permitted riding arrangement. Even equal sharing between that front bogie's axles would be 3,803.59 kg/axle in the stress case; actual sharing also depends on braking torque, suspension and twist. A catalogue selection using only gross/4 is inadequate.

Useful linear coefficients for the frame engineer:

- A 1 kN downward load at frame tip x=-47 produces +1.426 kN front reaction, −0.426 kN rear reaction and 2.337 kNm hogging at the front bogie. Rear-tip results mirror these.

- A 1 kN load at cabin edge x=-3 produces 1.219 kNm cantilever moment at the front bogie.

- A 1 kN/m uniform line load over the complete 312-inch cabin produces 3.019 kNm midspan sagging and 0.743 kNm support hogging in the ideal symmetric beam model.

Apply the approved platform crowd, stair, handrail and deployed lift reactions to the correct locations. These coefficients do not establish their required design loads. Deflection, local deck strength, lateral torsional buckling, crossmember twist and bracket prying require separate checks.

## 3. Required combined load cases

The vehicle engineer must issue a load-case schedule and numerical factors before member design. Candidate standards are design-method references, not a declaration that European mainline standards govern this Nevada attraction. NEN's published scopes identify EN 12663-1 body structural analysis/testing and EN 13749 bogie verification/manufacturing quality; full texts and project applicability remain to be reviewed [FR07-08].

| Case | Loads and combinations to resolve | Principal checks |
|---|---|---|
| V1/V2 | Tare, full nominal/stress payload, equipment at actual locations | Global bending, seat/floor anchors, individual wheel loads |
| V3 | Partial seating, passengers concentrated at one end/side, wheelchair/lift positions | Unequal reactions, lateral CG, loading stability |
| L1/L2 | Traction and braking with vertical load; both coupling tension and compression | Sill and draft-box stress, bolster/pivot force, local buckling |
| L3 | Reversing, slack run-in/run-out and emergency brake imbalance | Transient longitudinal forces; compression stability |
| C1 | Curve lateral acceleration, cant/crosslevel, wind, offset occupancy | Roll, wheel unloading, lateral rail force and retention |
| T1 | Diagonal track twist combined with vertical and lateral load | Suspension equalization, frame torsion, lowest wheel load |
| P1 | Stationary stairs/platform crowd, lift deployed and eccentric | Cantilever bending, local torsion, vehicle restraint |
| M1 | Jacking, lifting, rerailing and towing with designated points | Local reinforcement, restraint, permissible support sequence |
| F1 | Repeated joint impacts, braking, curves, loading and corrosion allowances | Fatigue hot spots and inspection access |

Do not add the entire train's coupling force to every axle, or omit it from the body frame. Coupling forces pass through longitudinal members; wheel braking reactions enter through brake mounts and bogie-to-body connections. Use the braking/coupling analysis to determine the force at each connection. A 65-ton train screen does not define each coach's design mass.

## 4. Gauge, CG and curve compatibility

The 110-inch exterior width is 3.06 times the 36-inch gauge. Width alone does not establish stability. A complete model must resolve sprung/unsprung mass, suspension roll centre and stiffness, damping, side-bearing geometry, wheel/rail forces, wind and track crosslevel. Do not claim a rigid tipping calculation proves derailment safety.

For a transparent sensitivity, place 16 stress-mass passengers on the eight benches on one side, with one centred crew member and symmetric 7,040 kg tare. Total mass is 9,505 kg; seat centres y=32 inches give transverse CG offset e=0.1984 m. At 7.5 mph on a 30 m radius, $v^2/(Rg)=0.03821$. With assumed CG height h=1.5 m and 0.9144 m support-spacing proxy, rigid inner-side weight fraction is $0.5-(e+h\cdot a/g)/s=$**0.2204**, versus 0.5 for centred stationary loading. At h=1.2/1.8 m it is 0.2329/0.2078. These are side totals, not individual-wheel predictions. Actual contact spacing differs from gauge; suspension, adverse cant, wind and twist are omitted. No acceptance threshold or safe speed is inferred. This demonstrates why partial loads and one-sided boarding deserve explicit assessment.

Using the 216-inch bogie centres, rigid body centre-chord offsets are 126, 75 and 50 mm for 30, 50 and 75 m radii. A 400×110-inch rectangular envelope then reaches approximately **1.681/1.572/1.516 m outside track centre** and **1.523/1.472/1.447 m inside**, respectively. These are preliminary geometric envelopes excluding stairs, deflection, tolerances, reverse curves and transition effects. Wheelbase, yaw stops and bogie sweep remain unknown; therefore no minimum curve radius is approved. Severn Lamb's identified Jupiter publishes 914 mm gauge and 50 m radius, but those figures describe its train, not this chassis [FR02].

## 5. Materials and real supply routes

The preferred steel underframe with light aluminium body/framing is an engineering development choice for repairability and separation of heavy train forces from the cabin. It is not a claim that steel is always lighter or cheaper. An all-aluminium underframe is a valid competing engineered proposal. Compare completed structure mass, stiffness, fatigue life, joints, corrosion protection and fabrication capability, not density alone.

Hydro documents 6005A extrusions for rail components, density 0.098 lb/in³, and temper/thickness-dependent properties; direct dissimilar-metal contact can cause galvanic corrosion [FR05]. These are supplier material data, not welded-member allowable stresses. Aluminium Association guidance explicitly treats weld-affected strengths and quality control [FR06]. Isolate aluminium/steel interfaces, retain engineered bonding paths for electrical safety, drain trapped water and provide replaceable exterior panels. Material grade, temper, weld procedure and heat-affected-zone design remain unresolved.

| Candidate | Verified primary-source evidence | Procurement limitation |
|---|---|---|
| Alan Keef | Current manufacturer describes passenger carriages, proprietary bogies and bespoke narrow-gauge work; 2024 weather-protected coach project included accessible provision [FR01] | No public rating/outline for this 914.4 mm, roughly 12 t coach; obtain a specific integrated proposal |
| Severn Lamb Jupiter | Manufacturer lists 914 mm gauge, 50 m minimum radius, 1% maximum gradient, enclosed option [FR02] | Whole-train reference; standalone truck availability and our weight/CG compatibility unverified |
| GHH-BONATRANS | Manufacturer designs wheelsets for customer bogie, brake and drive interfaces; provides calculation/test-backed development [FR03] | Exact gauge, wheel/axle design, order quantity, lead time and price unknown |
| Schaeffler FAG | Railway axleboxes and factory-sealed/greased TAROL/cylindrical units; application-specific housing and bearing validation [FR04] | No selected bearing part number or C/C0 rating; supplier must select for actual radial/axial duty, seals and temperature |

Public pages checked 17 October 2026 establish capabilities, not stock availability, a quotation or approval. Buying engineered running gear reduces integration risk but does not transfer responsibility for the coach/frame/track system. Fabricating bogies requires the same design evidence plus qualified manufacture and prototype verification; apparent simplicity is not an exemption.

## 6. Supplier qualification, assembly and release gates

A running-gear proposal must return a dimensioned interface drawing and load schedule covering: exact gauge and tolerances; wheel diameter/new-worn profile; wheelset back-to-back; flange thickness/height and root radius; rail-head match, cant and turnout/check-rail interfaces; axle fatigue calculations and material traceability; bearing selection/life, axial retention and seals; allowable static/dynamic vertical, lateral, longitudinal and roll loads; suspension load/deflection, bump/rebound travel, equalization and damping; bogie wheelbase and full yaw/pitch/roll envelope; brake mounting/thermal limits; bolster/centre-bearing/side-bearing interfaces; positive body retention, jacking/rerailing, inspection and replacement limits. Require unloaded and loaded wheel-load predictions and a permitted CG envelope. Obtain separate static, fatigue and exceptional-load definitions rather than a single ambiguous “ton” capacity.

The track and vehicle engineers jointly own the wheel/rail interface. Freeze neither profile nor turnout geometry until both sign the same controlled drawing. Route grade, radius, reverse-curve spacing, vertical transitions, crosslevel/twist tolerances and operating speed are required before final running-gear acceptance, even though the North Pole location can be supplied later.

Proposed execution sequence, subject to reviewed drawings:

1. Agree design basis, responsibility split and verification plan; approve load cases and choose the running-gear route.

2. Receive supplier CAD and mass/CG data; solve floor-height and suspension packaging; complete frame stress, fatigue, deflection and connection calculations.

3. Release fabrication drawings, material schedule, welding procedures, tolerances and inspection plan. Fabricate a frame on a surveyed fixture with accessible reference datums.

4. Inspect materials and fit-up before closing boxes; control weld sequence and distortion; perform specified weld examination; measure bolster coplanarity and diagonals before coating.

5. Install complete supplier wheelsets/bearings/brakes without unapproved machining or welding. Record assembly torque/retention, profile and back-to-back measurements; fit body and equipment.

6. Weigh every wheel empty and with controlled test loads; establish CG, clearances, suspension travel and loading reactions. Conduct engineer-approved static proof, twist, curve, brake and retention tests before passenger commissioning.

Farm crew can prepare fixtures, handle approved finishing and install noncritical trim under the issued work instructions. Specialist responsibility includes frame engineering, primary structural welding/examination, wheelset press fits, axle/bearing assembly, brake integration and acceptance testing. No pin diameters, member sizes, weld sizes or numerical fabrication tolerances are released by this report.

# 05 | Brakes and inter-car connections

Revision PF-TRAIN-S3A-R0 • 17 October 2026 • Preliminary engineering definition. **Not for fabrication, construction, passenger operation or procurement release.**

**Owner update:** track is to be laid at **0% longitudinal grade**. This is the governing project basis. The retained 1/2/3% cases are off-design sensitivities only; they are not route requirements. Final as-built profile and acceptance tolerance remain an engineering hold.

## 1. Recommendation and evidence

Retain the owner's hydraulic service-brake preference. Develop a distributed railway hydraulic system with brakes on every vehicle, a separately justified emergency function, and mechanically sustained parking on each coach. Advance it alongside a conventional automatic-air alternative until an experienced rail-brake integrator accepts responsibility for the complete system. A spring-applied caliper is a component; its presence does not establish a fail-safe train.

Hydraulic rail equipment is real, rather than an automotive improvisation. Knorr-Bremse lists HydroAct HC1P50 spring-actuated calipers with two hydraulic release circuits, force up to 100 kN per disc, 100 mm maximum disc width and approximately 60 kg without bracket. Its smaller HSC1P10N listing gives 20 kN, 45 mm disc width and 15 kg. These are manufacturer-listed examples, not selected components; the exact meaning of force, torque conversion, release pressure, friction pairing and operating duty require a controlled application drawing. [BC01]

Knorr-Bremse HydroControl Smart type u3HU is a railway hydraulic supply/control family for pressure- or spring-actuated calipers, listed at 0-100 bar; its accumulator configuration can support wheel-slide protection. HydroControl Basic HGK is a different variant without an accumulator. Neither establishes the complete architecture needed here. [BC02]

An industrial comparison shows why a rating alone is insufficient. Twiflex DS2610 lists GMXSH/XSH 7.5 at 11.2 kN **tangential disc force**, 65 bar minimum full-retraction pressure and 120 bar maximum pressure; its 13.62 kg caliper/thruster listing and force assume conditioned pads at nominal friction 0.4. The associated GMX family requires at least a 610 mm disc diameter. Packaging, mounting and rail-vehicle qualification remain unproved; do not order it for these coaches. The manufacturer-hosted sheet is dated 11/11, so a current controlled drawing is needed. [BC03, BC04]

Prefer an engineered railway drawbar between seldom-separated coaches, with a compatible detachable locomotive interface. Dellner's People Mover semi-permanent assembly uses a central double-acting damper, pivot anchor and cable support; its public page does not supply a rating or dimensions. Its brochure illustrates socket joints, draft gear and central connections. Voith's 430/530 railway coupling family is advertised up to 300 kN in tension and compression, while semi-permanent configurations can incorporate energy absorption. These establish actual railway principles, not small-train fit, price, lead time or attachment adequacy. [BC05-BC07]

## 2. Architecture for engineering development

**H1, preferred investigation:** each coach receives local hydraulic power/control equipment, separately isolated service zones for front and rear bogies, and a separately validated spring-applied/hydraulically released emergency/parking path. Local brake energy and control must permit a detached coach to brake and hold without the locomotive. The final design may use integrated railway actuators, but shared discs, brackets, fluid, electronics and release circuits are common-cause items that must be accounted for. No specific valve manifold is selected.

Driver service demand is distributed to local regulators; local pressure and actuator-state feedback return to a train brake controller. Brake-force balance must address empty, full, asymmetric and mobility-device

loading, not merely equal pressure on every car. A monitored hardwired emergency continuity function is separate from the routine command/data bus. Driver/crew emergency controls, breakaway detection and defined power faults demand a local emergency response. Stored springs supply application energy only if release pressure can escape and the mechanism remains operable. Trapped pressure, a blocked return or a stuck valve can defeat that intention.

Traction permission requires the defined doors/gates and boarding equipment secured, emergency circuit healthy, parking/emergency release confirmed, and valid brake-system status. A stop indication alone does not prove brake holding. Releasing an emergency demand requires a deliberate reset and checks; restoration of power must not cause automatic movement. Manual auxiliary release requires positive vehicle securing, a recorded isolation state and traction lockout until reinstatement. Interlocks must not prevent required emergency egress.

**A1, comparison:** automatic railway air service/emergency with a continuous brake pipe, local reservoirs/distributors, suitable actuators, and independent mechanical/spring parking. Wabtec's emergency valve vents the brake pipe to initiate quick action; its 26C controls brake-pipe pressure, and rail distributors translate this into local brake action. [BC08] This alternative has an established separation-response principle and may integrate more readily with a purchased locomotive. It adds compressor/dryer/reservoir installation and demands demonstrated propagation, recharge and leakage performance. Air pressure alone cannot be credited indefinitely for parking.

**Reject as the sole system:** one automotive-style master cylinder and long interconnected hydraulic line, locomotive-only brakes, an ordinary trailer surge brake, or emergency braking dependent solely on a consumer wireless link. These do not provide the documented car-level fault containment needed for this design. This is a project engineering decision, not a claim that a particular statute bans every such device.

## 3. Fault matrix and proof obligations

| Event | Required response to develop | Verification before acceptance |
|---|---|---|
| Normal service request | Coordinated proportional braking on all cars; monitor response and traction removal | Empty/full/in-between loads; application and release times at each bogie |
| Driver emergency or crew emergency | Traction removed; validated emergency application throughout train | Every initiating control, worst voltage/temperature and maximum load |
| Loss of locomotive or hotel power | Local emergency response and sustained parking remain available | Cut each source separately; show no shared-power single failure defeats braking |
| Loss of local controller power | Defined local application and train fault indication | De-energize one car; verify resulting stop and coupler loads |
| Service hose rupture | Contain loss to defined zone; remaining service/emergency path responds | Controlled fault injection, fluid-loss and pressure-decay measurements |
| Release hose rupture/low release pressure | Spring force may apply; prevent uncontrolled lock-up or unequal braking | Measured torque buildup and wheel-slide behavior; cold/hot/worn parts |
| Blocked return or trapped release pressure | Detect failure to apply; alternate validated path must stop train | Stuck-valve/blockage analysis and representative bench fault tests |
| Routine bus loss/stale command | Time-bounded safe response; independent emergency function remains | Broken wires, frozen outputs and invalid messages |
| Emergency loop open, short or cross-feed | Detect intended fault set; absence of an open circuit alone is insufficient | Open/short/cross-feed tests; common-cause review |
| Coupling separation | Detached portion brakes and holds locally; locomotive also reacts | Controlled separation simulation, no passengers; prove connector sequence |
| One car or bogie unbraked | Detect; meet approved degraded stopping condition or prohibit operation | Residual stopping, adhesion and longitudinal-load analysis |
| Brake seized on/pressure sensor misleading | Detect drag/overheat or inconsistent feedback; controlled stop | Fault tests independent of one pressure indication |
| Manual release left active | Departure inhibited and conspicuous isolation indication | Recovery/towing procedure and restoration check |

| Event | Required response to develop | Verification before acceptance |
|---|---|---|
| Parking and securing | Each detached vehicle holds under approved load, actual profile and environment | Independent hold test, endurance/leakage observations, securing procedure |

The table states requirements to prove, not demonstrated responses. Per-car independence must include connectors, power protection, fluid isolation, valve states and software/common wiring failures. A brake integrator must assign failure rates, diagnostic coverage, redundancy and acceptance criteria using the agreed governing standard. Farm crew should not prototype safety control logic.

## 4. Level-track stopping, adhesion and heat screening

The governing longitudinal grade is **0%**, per the owner's latest direction. The saved model retains 96 combinations across 55/60/65 tonnes, 5/7.5 mph, 1.5/3 s equivalent delay and service/emergency specific brake forces **0.50/0.80 m/s²**. Its 24 zero-grade cases are project-basis screens; the other 72 at 1/2/3% are off-design sensitivities. Delays and specific forces are analyst inputs, not standards, demonstrated performance or approved passenger decelerations.

Brake force B = M b. On level track, v1 = v, delay travel = vt, braking travel = $v^2/(2b)$, and total stop is their sum. In the retained nonzero-grade model, v1 = v + gst, delay travel = vt + $gst^2/2$ and braking travel = $v1^2/[2(b - gs)]$, with g = 9.80665 m/s². This preserves gravity acceleration during delay. Here b means force per mass **before grade**, differing from the previous Stage 2 net-deceleration target. At the new zero-grade basis those values coincide. Off-design downhill comparisons require normalized assumptions.

No rolling-resistance or engine-braking credit is taken. Rotating inertia, jerk, brake buildup shape, fade, wheel-slide cycles and curves remain to be added. These translational results are **not a conservative upper bound**. The as-built longitudinal profile must be surveyed; the responsible engineer must set acceptable deviations and their treatment. No numerical grade tolerance is invented.

| Speed | Downgrade | Service stop, 3 s delay | Emergency stop, 3 s delay |
|---|---|---|---|
| 5 mph | 0% | 11.7 m / 38.4 ft | 9.8 m / 32.2 ft |
| 7.5 mph | 0% | 21.3 m / 69.9 ft | 17.1 m / 56.1 ft |

Distances are mass-independent only because the assumed brake force scales directly with mass. At 55/60/65 tonnes, assumed service forces are 27.5/30.0/32.5 kN; emergency forces are 44/48/52 kN. Fixed hardware torque does not behave this way.

At **65 tonnes, 7.5 mph and 0% grade**, both ideal stops dissipate **0.365 MJ** of initial translational kinetic energy. The gravity-related continuous heat component is **zero**. Repeated-stop frequency, rotor/pad limits, hot starts, cooling and any dragging brake still govern thermal design. Manufacturer guidance distinguishes holding, dynamic and continuous duty. [BC09] The retained 3% cases previously produced 1.416/0.922 MJ per service/emergency stop and 64.1 kW continuous descent heat; these are explicitly **off-design examples, not project equipment loads**.

The following level-track emergency-command screens retain 65 t, 7.5 mph and 3 s delay. Failed-unit torque is not redistributed; unaffected brakes retain b = 0.80 for their own allocated mass. Adequate adhesion is assumed except the locomotive-only row, which explicitly limits force by assumed μ = 0.15. These are fault-study outputs, not permission to operate with a defect.

| Level-track case | Total stopping distance |
|---|---|
| All vehicles braking | 17.08 m / 56.1 ft |
| Accessible coach unbraked | 18.51 m / 60.7 ft |

| Level-track case | Total stopping distance |
|---|---|
| One regular coach unbraked | 18.87 m / 61.9 ft |
| One regular bogie unbraked, equal bogie allocation | 17.88 m / 58.6 ft |
| Locomotive unbraked | 19.10 m / 62.7 ft |
| Only locomotive braking at assumed adhesion0.15 | 27.19 m / 89.2 ft |

Wheel/rail adhesion bounds B by μgM_braked. Fully braked weight requires ideal μ of 0.0510 for b = 0.50 and 0.0816 for b = 0.80. The 60-case adhesion matrix uses assumed μ = 0.03/0.05/0.08/0.10/0.15 and braked-weight fraction = 1/0.75/0.25, with nonzero grades marked off-design. None is a measured site value. At level grade and all weight braked, assumed μ = 0.03 increases the emergency-command stop to about 29.2 m. It is an adhesion upper-bound screen assuming sufficient surviving hardware torque, distinct from the no-redistribution fault table. More clamp force cannot cure insufficient adhesion. Establish contamination controls, slide protection/sanding suitability and operating restrictions from tests.

At exactly zero nominal grade the gravity holding term is zero; **parking is still required on every detached vehicle**. Wind, inadvertent movement, actual residual grade, uneven wheel loads and securing margins remain to be evaluated. The retained 3% static figures (19.12 kN train, 3.87 kN regular coach, 3.23 kN accessible coach) are off-design sensitivities only. For brake sizing, convert wheel force to axle torque using actual loaded wheel radius and then disc torque using the caliper's friction definition; do not confuse clamp force with tangential force. Include equivalent rotational mass $\Sigma I/r^2$ in dynamic deceleration.

## 5. Connection loads and articulation

For the 65 t upper screen, assume 14.5 t locomotive and 50.5 t coaches distributed proportionally to existing stress coach allocations. These are calculation allocations, not a weighed consist. The governing **level-track** coupling launch screen uses 0.15 m/s² acceleration and analyst resistance coefficient 0.005. This differs deliberately from the propulsion launch study's 0.10 m/s² and 0.010 coefficient: they are distinct preliminary load cases, **not a unified design envelope**. The responsible engineer must reconcile the full resistance, acceleration and transient load combinations. Curve resistance is absent here.

| Joint | Trailing mass | Level start tension / reverse push compression | Loco-only braking compression* | Rear car unbraked compression** |
|---|---|---|---|---|
| Loco-R1 | 50.50 t | 10.05 kN | 16.57 kN | 1.96 kN |
| R1-R2 | 37.33 t | 7.43 kN | 12.25 kN | 3.74 kN |
| R2-R3 | 24.16 t | 4.81 kN | 7.93 kN | 5.52 kN |
| R3-Accessible | 10.99 t | 2.19 kN | 3.61 kN | 7.30 kN |

The reverse-push figures assume the same quasi-static acceleration and resistance magnitudes while the locomotive pushes the consist; they are not dynamic buff ratings. *Locomotive braking is limited by assumed μ = 0.15 on 14.5 t; this is a failure study. **Other vehicles apply b = 0.80; rear coach contributes zero. On uniform grade, joint tension C = B_trailing − M_trailing B_total/M_total; negative C is compression. This ideal brake-imbalance force is grade-independent because common gravity acceleration cancels. It

excludes differential grades, curve resistance and buildup. Perfectly mass-proportional braking gives zero quasi-static joint force, not zero transient force. Coaches braking with an unbraked locomotive give 9.01 kN head-joint tension. Nonzero-grade launch cases remain in a separately labeled off-design array.

These forces **must not become coupler design ratings**. Engineer separately sets buff/impact, reversing, collision, fatigue, rescue towing, vertical/lateral and proof loads; integrates draft-gear stroke; and evaluates compressive-force derailment on curves. A generic multiplication factor cannot replace that assessment.

Retain 30 in end-platform gap, 14 in pin setbacks and 58 in pin-center spacing only as sensitivity geometry. A 430 in equal-car center spacing gives approximate adjacent yaw 20.86°/12.52°/8.34° on 30/50/75 m same-radius curves; equal joint articulation is roughly half. This arc-spacing approximation does not cover reverse curves, turnout transitions, different locomotive geometry, vertical curves, suspension travel or twist. Perform an exact multibody swept-volume check with stops, wear, pin clearances, hoses, cables, stairs, guards and removable inspection covers. A single vertical pin alone does not supply all required pitch/roll freedom.

## 6. Detail release and inspection

The conceptual connection consists of a load-spreading frame anchor/draft gear, replaceable articulated bearing, retained pin/joint assembly, rail drawbar and the mirrored adjacent-car assembly. Provide positive primary retention and an independently reviewed secondary-retention concept; do not assume an ordinary safety chain will restrain a train or prevent derailment. No passenger passage across the coupler is approved.

For eventual pin engineering, average shear is $F/(n\pi d^2/4)$, projected bearing is $F/(dt)$, and bending stress is $32M_{pin}/(\pi d^3)$. Double shear requires verified load sharing; pin bending comes from actual contact widths and gaps. Check clevis net section, tear-out, prying, bearing fit, fretting, lubrication, retaining-device loads and corrosion. Drawbar checks include axial stress, eccentric bending, local/global buckling and approved fatigue assessment. Weld sizes, material grades, heat treatment, tolerances, pin diameter and plate thickness remain **HELD** pending calculations and controlled supplier data.

Release sequence: approve requirements and load cases; obtain interface/control drawings and duty ratings; complete failure analysis and independent review; build instrumented bench assemblies; test articulation and hydraulic faults; inspect traceable fabrication; conduct secured empty-vehicle trials, staged ballast tests and approved maximum-load stopping/holding tests. Record wheel loads, pressures, speeds, deceleration, coupler force, rotor temperatures and fault indications. Passenger service follows documented acceptance by the responsible professionals and authorities. Farm crew may maintain approved access/cleanliness and record inspections after training; safety-critical fabrication, hydraulic commissioning and acceptance require specialists.

# 06 | Propulsion, HVAC and electrical power

PF-TRAIN-S3A-R0 • 17 October 2026 • PRELIMINARY - NOT FOR FABRICATION OR CONSTRUCTION

## Recommended procurement boundary

Purchase a complete, supported locomotive/powered chassis from a railway vehicle integrator, with its controlled drivetrain, powered wheelsets, friction brakes and cooling package. Diesel-hydrostatic remains the leading study architecture; hotel generation remains separate. This preserves cabin space and avoids taking HVAC power from traction unexpectedly. The 14.5 t locomotive-and-hotel allocation includes generation and is **not verified adhesive mass**. Do not order individual drivetrain parts from this list. **Owner design basis: 0% longitudinal grade, with no intentional grades.** Survey must verify the as-built profile and the responsible engineer must set residual-grade and settlement tolerances; none is invented here. The former 1-3% cases remain off-design sensitivities only, not locomotive procurement requirements.

| Path | Practical consequence | Phase 3 disposition |
|---|---|---|
| Complete engineered diesel locomotive | Supplier owns tractive effort, cooling, controls, reversing and wheelset interfaces | Preferred; evaluate custom-coach mass, not advertised passengers |
| Used locomotive plus specialist rebuilding | Condition, braking, visibility, emissions, gearing and parts can dominate conversion | Alternate only after inspection and documented rebuild scope |
| New diesel-mecha nical/powershift | Efficient geared operation; converter crawl heat, reversing duty and rail final drives require matching | Retain alternate; no approved transmission application yet |
| New diesel-hydrostatic | Good low-speed control and reversing; substantial heat rejection at low road speed | Leading architecture for integrator proposal |
| New diesel-electric | Controlled axle torque and flexible packaging; added high-voltage, inverter and cooling integration | Credible specialist alternative, not farm-crew electrical fabrication |

**Jupiter identification.** Severn Lamb's catalog JUPITER publishes 914 mm gauge, 160-320 passengers, up to four coaches, 1% maximum gradient, 50 m minimum radius and 16 km/h maximum speed; diesel/LPG and enclosed bodies are offered [PU01]. It publishes no mass, engine rating or tractive-effort curve on that page. These values do not rate our coaches. Western Train's separate Jupiter has 38-inch direct-drive wheels, tender power and air brakes, with a 24-inch photograph label [PU02]; its FAQ describes diesel/Eaton hydraulic propulsion and custom gauges including 36 inches [PU03]. Neither is a verified 36-inch offer for this consist. Severn Lamb's custom electric Dayton locomotive is a different engineered product, not a higher diesel-Jupiter rating [PU17].

## Governing level-track traction screening

Reproducible calculations: calculate_propulsion_utilities.py reads the frozen source within this Phase 3 package. propulsion_utilities_calculations.json separately identifies project level-track cases and retained off-design grade sensitivities. Assumptions: g=9.80665 m/s²; 7.5 mph=3.3528 m/s; rolling resistance=0.005 of weight; starting resistance=0.010. These are sensitivity inputs, not measured performance. Curve and wind resistance remain **unquantified and excluded**. Starting and steady-running requirements remain nonzero on level track. The 65-t whole-train sensitivity does not increase any coach/frame structural allowable.

F_run=M g×0.005; F_start=M g×0.010; acceleration adds M a; P_wheel=F_run v. The table omits acceleration. Engine screen=P_wheel/0.75+10 kW assumed traction auxiliaries; hotel generation is

separate. Saved calculations also test 0.70/0.85 efficiencies, not guaranteed architecture performance.

| Mass t | Project grade | Run kN | Start kN | Wheel kW at 7.5 mph | Engine kW |
|---|---|---|---|---|---|
| 55 | 0% | 2.70 | 5.39 | 9.0 | 22.1 |
| 60 | 0% | 2.94 | 5.88 | 9.9 | 23.2 |
| 65 | 0% | 3.19 | 6.37 | 10.7 | 24.2 |

**At 65 t on level track, starting requires 6.37 kN before acceleration and curves.** Adding the study acceleration of 0.10 m/s² raises the starting force to **12.87 kN**. Once rolling at that acceleration, force is 9.69 kN, wheel power at 7.5 mph is 32.48 kW, and the same engine allowance is 53.31 kW. Starting force at zero speed must not be confused with continuous power at travel speed. Acceleration is a screening assumption, not an approved passenger operating limit.

| Assumed adhesion coefficient | Driven mass for 65-t start, no acceleration | Driven mass for start plus 0.10 m/s² |
|---|---|---|
| 0.10 | 6.50 t | 13.13 t |
| 0.15 | 4.33 t | 8.75 t |
| 0.20 | 3.25 t | 6.56 t |

The 14.5 t locomotive-and-hotel allocation is not measured driven mass: the generator, tender, nonpowered axles and fuel distribution may reduce adhesive weight. Level operation substantially reduces the traction requirement, but wheel loads, CG, contamination, starting on curves and braking still govern integration. Verify actual resistance and adhesion; do not automatically downsize the engine from this screening calculation. Gearing and wheel torque, auxiliary demand, cooling and transient power require supplier confirmation.

The saved 1-3% cases are retained **only as off-design sensitivity/history**. They do not define the planned railway or override the owner's zero-grade instruction. Any measured residual longitudinal grade must be compared with engineer-approved as-built tolerances and the integrated vehicle's verified performance. Drainage must be designed alongside the level track without silently introducing a track grade.

## Real components and matching limits

| Candidate | Documented evidence | Matching decision |
|---|---|---|
| Cummins F3.8 154 | 115 kW; peak 620 Nm at 1600 rpm [PU04]. Cummins rail guide lists 115 kW at 2200 and 2500 rpm for intermodal/maintenance use [PU05] | Investigate 2200-rpm calibration or a lower appropriate rating against level-track duty. Not a locomotive emissions/application approval. The 115-kW example is not a required or selected rating |
| Danfoss H1P060 + H1B080 | Pump 60.4 cc/rev, 3500 rpm rated, 420 bar maximum working; motor 16-80 cc, 1.27 Nm/bar theoretical, 3200 rpm rated at maximum displacement [PU06-07] | Real hydrostatic chain candidate; OEM to select controls, topology, reductions and duty. Independent brakes required |
| Allison 1000/2000 | Family published 154-272 kW and up to 950 Nm [PU08] | Family maxima do not select a model or establish rail duty. Application engineering, full reverse/crawl thermal data and exact rating required before adoption |
| Danfoss EM-PMI240-T180-4400 | Class F at 65°C coolant: 80 kW, 175 Nm continuous, 4400 rpm nominal; nominal 500 VAC, inverter supplied. Minimum coolant 20 L/min; derate above 65°C [PU09] | Published power exceeds the present level-track screen; actual low-speed torque, gearing and duty maps still govern. Two motors can distribute torque but do not double diesel power |
| Danfoss EC-C1200-450+MC180 | 150 kVA/180 A nominal; 750 VDC nominal link; liquid cooling up to 65°C, minimum 10 L/min [PU10] | Candidate motor controller; supplier must match actual bus, motor map and controls. Do not adopt MC350 peak claims: that version requires coolant ≤40°C |
| Gmeinder GGT190EHA/497 +190E/327 | Railway axle-gearbox reference, 12,150 Nm maximum input, 2720 rpm, ratios 4.97/3.27, for 760-mm locomotive [PU11] | Specialist capability reference only; not a 914-mm drop-in axle or selected ratio |

Hydrostatic example at 2200 rpm: H1P060 theoretical flow=132.88 L/min; at 420 bar, ideal hydraulic power=93.02 kW. Two 80-cc motors in parallel would each turn 830.5 rpm and develop theoretical 533.4 Nm before losses. This demonstrates a coherent scale, not continuous operating approval. Required reduction is F×wheel radius/(motor count×actual torque×gear efficiency); wheel diameter and actual efficiency remain hold points. The 420-bar value is the pump's maximum working limit, **not a prescribed setting**. Case drain, charge pressure, filtration, overspeed, torque limiting and measured cooler duty require manufacturer calculation.

A diesel-electric proposal needs diesel→matched generator/rectifier→protected DC link→inverters→motors→rail reductions. The motor datasheet alone does not supply that system. No generator/inverter/brake combination is represented as matched or available from stock.

## Thermal and hotel power definition

Retain 33 thermal occupants in each regular car and 14 in the accessible car; 7.5 L/s/person ventilation remains a **study assumption requiring an applicable basis**, not a code claim. Total outdoor air=0.8475 $m^3/s$ before doors. No ventilation reduction is used to make equipment fit. Summer 110°F outside/75°F inside; winter 30°F/68°F. Clear shell 7.7724×2.6416×2.2 m; glass 9 $m^2$; opaque U=0.7, glass U=2.7 $W/m^2K$; SHGC=0.5; glass incident solar=500 $W/m^2$; additional roof sol-air=20 K. These are test inputs, not purchased material properties or local design-weather verification.

| Coach | Dry cooling incl. 15% kW | Humid stress incl. 15% kW | Winter loss before gains/warmup kW |
|---|---|---|---|
| Regular, 33 people | 16.96 | 22.70 | 8.74 |
| Accessible, 14 people | 10.49 | 13.28 | 5.10 |
| Three regular + accessible | 61.35 | 81.39 | 31.31 |

Humid stress adds outside-minus-inside humidity ratio 0.006 kg/kg to **both ventilation and 0.03 m³/s per-car average door/leakage air**. Passenger total=120 W/person, including 45 W latent. Door-opening volume/time, North Pole dwell with doors open, duct pressure, window performance and thermal bridges need refinement. The station location does not block this work. Keep compressor input and thermal cooling distinct.

At assumed delivered COP 2.5 plus 3 kW other electrical demand, train running demand is 27.54 kW dry/35.56 kW humid; at COP 2.0 it becomes 33.68/43.69 kW. Four assumed 10-kW resistance heaters plus other loads need 43 kW. Actual fan power must not be double-counted; measured compressor starts/drive harmonics govern transient sizing.

Northwest Rail Electric NW-64-CN is a roughly five-ton condenser for rail/tourist applications [PU13]. Its nominal ~17.6 kW does not establish delivered high-ambient capacity; a matched evaporator/ventilation/heater package and regular-car humid capacity are unresolved. Konvekta HVAC 5401 publishes 16 kW at ta=ti=40°C, 7.2 kW heating, 400 VAC and 275 kg (240 kg optional) [PU12]. Neither rating condition matches 110°F/75°F. One 5401 is not demonstrated to satisfy the regular car, and 400V/60Hz compatibility is unverified. Obtain high-ambient sensible/latent output, compressor input/start data and current US configuration before selection.

Stadco 6-42T4F19 publishes 42/40/38 kW intermittent/prime/continuous at 240/480V three-phase, 60Hz; 6-80T4F19 gives 80/74/70 kW [PU14]. Reference conditions are 80°F and 500 ft; vendor derating is required. Published approximate envelopes/masses are 72×32×30 in/1790 lb and 103×36×32 in/3126 lb. The 42 model is marginal against humid/heat cases before derating; the 80 model warrants a site-rated offer. The proposed 50 model remains under development [PU15]. Keep the **40-70 kW delivered-continuous investigation envelope**; no generator is selected. Separate source and load step analysis follows manufacturer application guidance [PU16].

## Installation interfaces and release holds

E01 conceptual single-line: isolated shore inlet and diesel genset → mechanically/electrically interlocked break-before-make transfer → main isolation/protection → protected articulated trainline → each coach lockable disconnect → HVAC/heating, lighting and low-voltage chargers. Study 480V three-phase/60Hz; retain 240V alternative. No paralleling. Shore-connected interlock inhibits movement. Neutral switching and the one appropriate neutral-earth bond arrangement require the responsible engineer's source/earthing design; do not bond neutrals in every car. Dedicated protective-earth continuity bonds vehicles; couplers/rails are not protective conductors. Trainline connectors need dead-before-disconnect interlocks, environmental rating, strain relief and articulation checks.

Each coach has independent battery-backed emergency lighting, communications and specified essential controls. Preliminary energy formula: Ah=P×hours/(V×efficiency×usable discharge×end-of-life factor×temperature factor). An illustrative 150 W for 2 h at 24 V, 0.90 efficiency and three 0.80 factors gives

27.13 Ah. This is **not** a selected battery or mandated duration; brake/door energy, surge loads, chemistry and approved rescue time remain unresolved. Brake power/failure behavior must be defined independently of hotel AC.

Before wiring schedules: approve topology, available fault current, protective coordination, earth-fault behavior, cable length/ambient/grouping/flex duty, phase balance, voltage drop, connector rating, HVAC starting kVA, and manufacturer minimum circuit/maximum protection data. Do not guess breakers or wire sizes.

Before locomotive fabrication: supplier-certified tractive-effort/speed maps, powered axle loads, reductions, failure behavior, emissions application, fuel map and cooling/exhaust installation limits. Fuel capacity derives from actual duty consumption plus approved reserve; no guessed gallon tank. Radiator/oil-cooler intake and discharge must work at standstill in dust under the decorative shell; provide service withdrawal paths. Keep both diesel exhausts and heat away from occupied-air intakes and queues in stationary/reverse winds. Verify sound/vibration, aftertreatment heat, fire separation, condensate discharge and inspection access. Qualified vehicle/HVAC/electrical engineers and the integrator complete these designs; farm crew can fit approved finishes and supports after interface release.

Source register PU01-PU17, URLs, dates and limitations are in propulsion_utilities.json. All accessed 2026-10-17; no contacts, stock confirmation, quotations or orders made.

# 07 | Bodies, interior, stairs and accessible boarding

**PF-TRAIN-S3A-R0 · 17 October 2026 · Preliminary; not for fabrication or construction.** This package develops a buildable arrangement for review. It does not release structural sections, fasteners, welds or an accessible rescue system. North Pole location remains deferred; passengers disembark there, so both stops need the same boarding capability. Ride programming and business projections are excluded.

## 1. Arrangement to develop

Retain the 312 × 110 in external cabin, 306 × 104 in clear floor, 3 in nominal wall envelope and 44 in end platforms. The 28 in rail-to-floor dimension remains a packaging assumption. Regular cars retain eight transverse rows, two 40 in two-person benches per row, 32 in pitch and 24 in aisle. Bench depth is a 20 in envelope; the approximately 12 in unoccupied pitch gap is not demonstrated knee clearance. Use warm wood-look removable liners over a separately engineered body frame; decorative panels do not carry passenger, guard, roof-equipment or coupling loads.

Retain 11 fixed seats and two 60 × 36 in reserved wheelchair bays in the accessible coach. The continuous 36 in route, front 60 in turning area and two companion seats must survive actual seat backs, footrests, belts, heater guards and door hardware. The existing aisle is offset: do not silently substitute a centered aisle. End doors remain 32 in clear regular / 36 in accessible; side bridge doors remain 42 in clear. These are reservations, not proof of accessibility or evacuation compliance.

Use an engineered steel underframe with aluminum body modules as the development baseline; obtain an all-aluminum engineered body/frame alternative. This gives the farm a repeatable cladding/interior assembly while specialists control fatigue-critical primary structure. Separate aluminum from steel with an engineered insulating interface, sleeves and sealed drainage paths; separately provide deliberate electrical bonding straps. Isolation must not defeat fault-current bonding.

## 2. Floor, wall and roof assembly options

| Layer/system | Development arrangement | Evidence and release condition |
|---|---|---|
| Floor | Replaceable supported panels above underframe, hard inserts at seats/lift/securement, sealed edges, continuous washable finish | Compare Coosa Bluewater 26 and supplier-designed Plascore aluminum honeycomb. Panel thickness/span and point loads require calculation and testing; no screw into foam/honeycomb alone for seats or restraints. |
| Wall | Exterior coated aluminum sheet or factory sandwich module; drained window surrounds; continuous thermal break; removable interior liner | Outer sheet gauge and post spacing remain structural decisions. Mechanical retention supplements a qualified adhesive system where consequences warrant. Keep the 3 in envelope provisional. |
| Insulation | Rail-specific closed-cell insulation with sealed seams; maintain access and drainage | ArmaFlex Rail SD is a candidate, not a fire-approved wall assembly. Its published US rail test claim is thickness-specific, 3-13 mm. Do not extrapolate to a 2 in stack. [BB06] |
| Roof | Curved framed modules with continuous weather layer and removable interior panels | Support HVAC through engineered framing, not roof skin. Keep penetrations above drainage paths; provide sealed curb, drip edge and service access. |
| Finish | Gerflor Tarabus Sirius NT trial flooring with coves and sealed seams | Published 2.25 ±0.15 mm, 2.20 ±0.15 kg/m²; declared slip/fire tests do not establish the complete train assembly's approval. Verify wet farm-soil cleaning performance. |
| Glazing/doors | Supplier-made retained safety glazing and vehicle-rated latches; mechanical inside emergency release independent of power | Glass type, retention, emergency opening size and number depend on approved hazard/code basis. Exterior slider/pocket arrangements need crush protection and weather trials. |

Material weight screening for the **221 ft² / 20.532 m² clear cabin floor only**: 3/4 in Bluewater 26 at nominal 26 lb/ft³ is 359 lb / 163 kg; 3/4 in Plascore AA3.6-80 at published typical 0.94 lb/ft² is 208 lb / 94 kg. These are equal-thickness mass comparisons, not equal-performance substitutions. Add floor covering approximately 45 kg before adhesive, inserts, perimeter closures, supports or waste. Neither total is a completed floor assembly. Plascore supplies open panel edges; edge closures and potted inserts need engineered shop details. [BB03, BB05, BB08]

Standard Coosa's FMVSS 302 claim is distinct from its separately ordered ASTM E84 Class A range. Neither automatically satisfies the applicable passenger-rail/amusement fire, smoke, toxicity or complete-assembly requirements. Obtain the exact ordered product's certificates. Sika Sikaflex-252 is a candidate vehicle panel adhesive, with 10-35°C application range; actual painted substrates need adhesion trials, bondline design and recorded curing. Do not use its typical lap-shear strength directly as a structural allowable. [BB04, BB07]

## 3. Boarding and steps

**Preferred normal boarding:** station platforms at the accessible side-door level at both stops, with captive rated bridge equipment, edge protection and staff-controlled gates. Establish rail-to-platform height only after tare/loaded suspension deflection, cross-level, wheel wear and floor depth are known. A bridge must accommodate those movements, not merely a nominal 28 in floor. At every boarding point immobilize the train, confirm permitted door side and verify the bridge/stair is deployed before opening passenger gates. Prove the stowed state before traction release.

For regular-car end platforms, develop a side-facing stair cassette at each usable end: retain the full longitudinal 44 in platform landing and shield the coupling below/behind a solid barrier. Do not route passengers across the drawbar gap. A **mockup**, not a released vehicle stair, uses 32 in clear stair width, 11 in going and four equal rises to a 28 in deck: 7 in/rise and three treads, producing 33 in outward run before

rails/nosing. Compare 36 in clear width during trials. A 36 in mockup handrail height is an ergonomic trial, not a governing code assertion.

Ground sensitivity: floor 28 in above rail gives 24, 28 or 36 in rise when landing ground is +4, 0 or −8 in relative to rail. Four risers become 6, 7 or 9 in respectively; the last is unacceptable as the retained 7 in study and requires redesign. Five rises at the 36 in case give 7.2 in and four 11 in treads require 44 in outward run. Thus a fixed stair sized from rail height alone is not a solution.

Prefer retractable/deployable stairs with positive stow lock, deployed support, guarded moving parts and state detection; make manual release accessible to staff after immobilization. Fixed stairs remain an alternative only after low-step and swept-envelope clearance at curves, crossings, turnouts and trackside obstructions. For either design calculate 3D sweep with suspension roll and maximum step projection. Rails, latch forces, structure and proof loads remain engineer-defined. A loose household step is not the boarding system.

## 4. Accessible boarding, securement and evacuation

If the attraction is classified under ADA amusement-ride provisions, the Access Board provides specific approach, gap and wheelchair rules; these are not ordinary stair dimensions transplanted onto a vehicle. Its direct gap benchmark is 3 in maximum horizontal and ±5/8 in vertical under normal passenger loading; otherwise the cited bridge/ramp provisions apply. Its side-entry advisory illustrates 42 in opening with 60 × 36 in maneuvering space. Check actual shoulder alignment and adjacency of the companion seats; their presence alone does not establish compliance. The final classification and any other applicable vehicle rules remain jurisdictional holds. [BB01, BB02]

**Ground access option:** BraunAbility NCL1000IB3754HB-2 has a published 1,000 lb lifting rating, 37 × 54 in clear platform, 385 lb mass, 48 in maximum travel and manual backup. Its June 2026 guide requires **45.85 in minimum clear doorway**, 67 in opening height and reserves 50.96 in overall lift width plus 18 in threshold depth. Therefore the current 42 in side door cannot accept it. Reserve an **optional, unintegrated 48 in clear lift-side opening** within the front boarding zone while retaining the opposite 42 in bridge opening; do not revise current CAD or assume 11 seats remain proven. Confirm mounting, guard/belt, stowed envelope, headroom, turn, bogie interference, 12 V/70 A maximum supply, special rail suitability and suspension roll with the supplier and engineer. Its 174.6 kg mass is within, not additional to, the existing 440 kg boarding/rescue allowance until reconciled. [BB09, BB10]

The explicitly rail-listed Ricon PowerLift is rated 300 kg; the 145 kg rider + 150 kg device stress case already reaches 295 kg. Its 5 kg nominal margin and unverified packaging make it an alternative, not the default. A lift’s rating does not approve the car’s mount or resistance to roll. [BB11]

Develop two complete Q’Straint QRT-360 four-point kits with lap/shoulder occupant belts as the securement candidate; manufacturer lists rail among applications. Keep anchor backing and belt loads connected to primary frame members; specify device compatibility, occupant orientation and emergency release with the responsible engineer. Wheelchair brakes alone are not a secured-travel solution. [BB12]

**Independent rescue concept:** carry a rated bridge/ramp interface and maintain a reachable, firm rescue access corridor for a separately rated mobile evacuation platform/lift. Reserve access to either side of the accessible coach; a blocked or failed onboard lift must not trap passengers. Survey later must identify where rescue equipment can position at every permitted train stop, its slopes, bearing area, reach and emergency responder approach. Validate a second side transfer path with the actual heaviest/largest admitted device, dead electrical supply and failed onboard lift. Until this is demonstrated, accessible off-station evacuation remains a no-passenger-operation hold. Do not substitute stairs, manual wheelchair carrying or an unplanned transfer to an evacuation chair as the universal solution.

## 5. Assembly and verification sequence

1. Freeze the load/interface schedule with the vehicle engineer and body supplier: bogies, frame twist, seat/guard forces, lift eccentric load, wheelchair anchors, roof equipment and electrical bonds.

2. Build a ground-level full-size interior mockup before cutting structural parts. Use adjustable walls/seats and the exact door/lift clearance templates.

3. Specialist fabricates and inspects the first underframe and body modules to released drawings; record material certificates, dimensional survey, weld procedures and inspections. Farm staff may prepare nonstructural templates and removable panels.

4. Dry-assemble modules; measure doorway diagonals, floor flatness, route clearances and body twist; install approved galvanic isolation and bonding. Photograph concealed structure and backing before closing.

5. Fit floor inserts/closures, drainage and wiring/condensate routes; prove access to all service points. Complete representative adhesive coupons before panel installation.

6. Farm crew can fit released liners, insulation, trim and finishes under documented supplier processes. Qualified installers fit glazing, critical doors, stairs, securements, lift and electrical systems.

7. Perform leak/drain tests before liners close; check fastener records, adhesive batch/cure conditions and all barriers. Weigh the finished empty car at each wheel; update tare/CG and loading limits.

8. Engineer-witnessed empty/ballasted structural and interface tests precede occupied mockup demonstrations. Test door/stair/lift faults, manual release, lighting loss and rescue. No human is a proof-test weight.

## 6. Mockup protocol and release evidence

Use six adults spanning body sizes: four seated and two passing/standing only for circulation; standing is not passenger capacity. Check 20 in seat allocation, coats, knee/foot clearances, reaching handholds, exiting each end and door-open conflicts. Record video, measured minimum clearances and obstructions; a comfortable-looking CAD model is not a pass.

In the accessible mockup use real manual/power chairs and a scooter; include both bays occupied, both companions seated, installed belt hardware, 36 in continuous route, 60 in turning maneuver and each boarding direction. Test normal lift/bridge use and the independent rescue path with trained personnel and safe mock equipment. Structural proof loads, allowable deflections, maximum emergency forces, evacuation performance criteria and test witnessing must be written by the responsible specialists before testing. Resolve failures by removing seats or changing geometry before any seat count is frozen.

# 08 | Track, stations, fabrication and acceptance

PF-TRAIN-S3A-R0 · 17 October 2026 · Preliminary engineering definition - not for fabrication or construction.

## Direction and release boundary

Proceed with vehicle engineering now. North Pole location will be supplied later; no site alignment is selected here. Guests will disembark there, so both origin and North Pole need normal and accessible boarding, safe train holding and accessible routes beyond the vehicle. Ride experience and business projections are outside this work. The companion JSON contains the expanded survey specification, interface fields, purchase prerequisites, inspection records and responsibility matrix.

Adopt the owner's 0% longitudinal track grade as the design target. Survey and as-built measurements must verify the resulting local grades; acceptable deviations and maintenance limits remain for the track/vehicle engineers to set. Level rails do not mean flat drainage: formation crossfall, ditches, outlets and culverts need independent civil design. Retain 36-inch gauge for evaluation, four coaches, 400-inch coach frame envelopes and 30-inch inter-coach gaps as the shared study basis. Rail section, wheel profile, allowable radius, turnout geometry and tolerance limits are unreleased. A gauge label alone cannot match a locomotive, bogie, turnout and track. ASTM's public F2960 scope expressly recognizes the need to coordinate subsystem interfaces [TD04].

## Responsible authorities and standards

Clark County's current 2026 ARPM code applies to installations of 30 days or longer unless exempt; operating only 20 evenings does not establish an exemption. The Building Official determines classification. It calls for Nevada design-professional responsibility, reviewed documents, manufacturing quality assurance, inspections, commissioning and an approved on-site Qualified Individual [TD01]. Obtain a project-specific classification and standards/edition matrix before safety-critical procurement. County building-code adoption is a separate facility-design consideration [TD06].

The FRA policy's exceptions include operations below 24-inch gauge and off-system, insular tourist operations. An in-use public road crossing, in-use at-grade rail crossing, certain bridges, or another railroad within 30 feet defeats the stated insularity test. A 36-inch train is not automatically exempt [TD02]. NRS 455B separately addresses permanent public amusement facilities and operator safety duties [TD03].

Disembarking at a separately visited North Pole makes the service-description question more consequential; it does not itself determine classification. Under 49 CFR 37.37, primarily recreational conveyances and private transportation services follow different accessibility paths [TD07]. Finalize the applicable vehicle requirements with a qualified specialist. Both stations still require an accessible boarding interface; off-station rescue remains a separate system.

Also coordinate Comprehensive Planning on existing SUP/parcels and use changes; County Fire Prevention and Moapa Valley Fire District on plan-review/response roles, fuel and rescue; and Public Works/civil reviewers on drainage, crossings and off-site interfaces [TD11-14]. Existing farm approvals have not been established as covering this railway.

Current public ASTM scope pages were reviewed, not licensed full standards. F2960-23 addresses permanent amusement track; F3054-26 addresses railway operations. Their exclusions include FRA jurisdiction in whole or part [TD04-05]. These are proposed references pending agreed applicability, not a claim that every clause or edition is County-adopted. No construction tolerance has been copied from an unseen standard. Nevada responsible-charge/seal requirements also apply to professional documents [TD08].

## Survey specification to issue when the owner marks candidate areas

Ask a Nevada land surveyor for a boundary/control/topographic scope covering candidate corridors, station areas, storage/shop and drainage catchments. A screenshot or consumer GPS trail is useful for briefing, not construction setting-out. Deliver:

1. Parcel/APN and title/easement reference schedule; resolved boundary and right-of-way geometry; control monument descriptions and photographs.

2. Explicit horizontal reference frame/epoch, projection, vertical datum, units and ground-to-grid method. Distinguish international feet from legacy U.S. survey feet. Deliver point CSV, coded feature list, breaklines/TIN in LandXML or agreed equivalent, editable DWG/DXF, and signed PDF survey.

3. Ground shots and breaklines sufficient to model ditches, channel banks/beds, embankments, slopes, irrigation, walls, roads, crossings and abrupt changes. Include overhead obstructions, structures, fences, trees, culvert ends/inverts/sizes and visible utilities with confidence/source labels. Underground service locations require separate investigation before excavation.

4. A documented horizontal/vertical accuracy and closure report. Surveyor and track/civil engineer must agree accuracy, point density, corridor extents and supplemental checks before fieldwork; do not substitute an arbitrary grid for actual breaklines. Final track geometry needs a separate precision setting-out and as-built plan.

5. Civil/geotechnical follow-up: bearing/settlement, fill, expansive/collapsible soils, erosion, irrigation saturation, drainage/flood exposure, compaction criteria and material testing. Scope investigation to the selected corridors; no assumed ballast section or culvert capacity.

## Track-vehicle interface control

Issue a signed interface drawing/table connecting the vehicle engineer, track designer and suppliers. It must specify rail profile/material/hardness and cant; wheel diameter/profile, back-to-back and flange geometry; gauge measurement plane and limits; turnout check gauge/flangeways; actual wheel and axle loads; speed, twist and crosslevel limits; suspension travel and wear states; curves, transitions and vertical curvature; coupler yaw/pitch/roll and swept stairs/body/utility envelope. Include the maintenance rejection limits as well as new-build limits.

Use 0% longitudinal grade for the main traction and braking calculation; retain 1/2/3% only as off-design sensitivity. Continue 30/50/75 m curve-radius studies; none is released as permissible track. Require clothoid/other transition treatment, reverse-curve separation and any local crest/sag geometry to be engineered with the actual bogie/coupler system. A short tangent between opposing curves is not automatically sufficient. Local grade, settlement and construction variation must close traction, braking, parking and recovery checks together. Zero nominal grade does not eliminate parking brakes or rollback protection. New and worn wheels, empty/asymmetric/full load and adverse track geometry belong in the clearance/dynamics analysis.

Route selection can later compare a continuous loop, terminal loops and reversing operation against surveyed constraints. Vehicle work now must preserve reversing/recovery requirements. Do not select a push operation until locomotive visibility, braking, coupler buff loads and train-control procedures are supported.

## Station envelope without a location

Four study coach envelopes and three gaps occupy:

**L_coaches = 4 × 400 + 3 × 30 = 1,690 in = 140 ft 10 in (42.926 m).**

This excludes the locomotive, locomotive-to-coach gap, outer coupling projections and end stopping margins. A full train berth is L_coaches + L_locomotive + G_locomotive + E_front + E_rear + any other projections not already included. Freeze each endpoint definition before using it on a plan. A coach-only boarding platform follows actual door/step/lift positions plus stopping and circulation margins; it need not equal the whole train berth. Neither length is yet released.

Develop the same station-interface schedule for both stops: rail datum; empty/full floor and suspension levels; boarding side; door/step/lift deployment; permissible gaps; swept clearance; stop-position tolerance; drainage; route gradient/crossfall; guarding/gates; lighting; emergency access; departure interlock; brake holding and rollback prevention. Level or near-level boarding is the preferred study; final height depends on the vehicle. If amusement-ride criteria apply, Access Board section 1002 addresses loading gaps and alternatives, but it does not establish this project's classification [TD09]. A locomotive berth, passenger platform and accessible rescue area are different envelopes.

## Farm-crew work packages and pilot

Maximize trained crew participation after reviewed plans and inspection gates. Crew can manage traceability, receiving, approved earthwork, drainage placement, tie/ballast handling, instructed fastening, finish work and records. The responsible designer sets the geometry, materials, procedures and acceptance limits. Qualified crew may perform specialist work only when the required qualifications, equipment and inspection arrangement are actually demonstrated.

Start with a specialist-led, isolated pilot using the selected rail/tie/fastening system: surveyed tangent, transition/curve segment, representative joint and controlled ballast/drainage section. Include a turnout or crossing only if it is part of the released pilot purpose. Length and curve geometry follow the test objective and stopping/runout analysis; no default pilot dimensions are asserted. Keep it isolated from guests and other equipment. Measure geometry before loading, under the defined test load where required, and after loading/settlement. Record corrective work and repeat measurements before expanding production.

Rail bending requires the approved section-specific tool and procedure; measure curvature rather than forcing rail against spikes. JENNMAR lists a #3H-25 bender for 25-60-pound rail, but the page has inconsistent general descriptions; obtain model instructions and written profile compatibility before selection [TD10]. Confirm units, rail grade and minimum bend radius. Driver, tamper, rail drill/cutter, lifting equipment and a calibrated 36-inch gauge/crosslevel tool remain procurement items, not approved products. A laser/total station establishes line/elevation; a track gauge checks gauge/crosslevel; a chord/offset method checks curve shape. None replaces the others.

## Sequence and evidence for release

1. **Definition:** appoint responsible engineers/inspection agency; agree jurisdiction, standards, interface ownership and document control. Prepare information requests; do not send or order without owner instruction.

2. **Prototype release:** close wheel/rail, running gear, coupler, brake and structural drawings; approve calculations, materials, fabrication procedures, inspection/test plan and controlled deviations.

3. **First article:** mock up boarding/end platforms; manufacture one chassis/coach with traveler records. Verify materials, weld qualifications/NDT, dimensions, wheel loads, retention and accessible equipment. Release repeat fabrication only after its relevant results pass.

4. **Track release:** accepted survey, geotechnical/drainage design, permitted alignment and signed interface precede bulk materials and earthworks. Procure a matched rail/joint/fastening/tie/turnout package; used rail requires condition and provenance acceptance.

5. **Integration:** geometry/as-built inspection, static load and clearance checks, instrumented speed/brake/holding tests, curves/twist, control faults, breakaway and no-power recovery. Use secured

ballast/manikins before passengers. Engineers set loads, limits, instruments and safe fault-injection method.

6. **Commissioning:** closed defects, approved manuals, trained staff, demonstrated accessible boarding/evacuation at both stops and worst rescue location, authority/insurer acceptance and required permits/certificates. A supplier's shop test does not release the completed railway.

The JSON matrix identifies each test's evidence and release owner. Final forces, welds, pin sizes, ballast/tie dimensions, rail-joint details, temperatures, tolerances and test acceptance thresholds remain explicitly held for the responsible professionals and OEMs.

# 09 | Coordinated interface register

The editable register accompanies the report. Close the interface with the responsible disciplines before releasing the connected fabrication or procurement package. A resolved dimension in one subsystem does not automatically release its neighbor.

| ID / interface | Definition and current basis | Owner / gate |
|---|---|---|
| IC01 Wheel/rail | Gauge, wheel profile, back-to-back, flange/flangeway/check gauge, turnout limits. 36 in nominal study gauge only | Vehicle + track engineers. Before rail/wheelset purchase |
| IC02 Bogie/frame | Bogie mount pattern, bolster/center bearing, side bearing, rotation, suspension travel, jacking. 216 in bogie centers provisional | Bogie supplier + structural engineer. Before frame sizing |
| IC03 Frame/body | Structural depth, wall/seat supports, underfloor services, corrosion isolation. 3 in old slab is a display envelope | Structural + body designers. Before floor elevation freeze |
| IC04 Vehicle/station | Loaded/unloaded floor height, suspension travel/roll, horizontal gap, bridge/lift reach. 28 in floor above rail provisional | Vehicle + civil/accessibility designers. Before platform and doorway fabrication |
| IC05 Draft system/frame | Coupler height, angular travel, buff/draft ratings, energy absorption, attachment load spread. 30/14/58 in connection sensitivity only | Coupler supplier + structural engineer. Before draft structure fabrication |
| IC06 Brake/running gear | Wheel/disc geometry, friction vs speed/temp/wetness, actuator mounts, per-axle effort. Hydraulic service candidate retained | Brake integrator + bogie supplier. Before wheelsets/brakes purchase |
| IC07 Brake/controls | Demand, emergency continuity, local power/pressure, separation response, status and traction inhibit. Functional architecture, no final safety circuit | Brake integrator + controls engineer. Before control design release |
| IC08 Locomotive/train | Driven mass, drawbar effort, coupling force, total mass, grade, curves, reverse duty. 55/60/65 t sensitivity envelope; owner longitudinal-grade target 0%, curves and actual driven mass unresolved | Locomotive supplier + vehicle engineer. Before locomotive selection |
| IC09 Hotel power/propulsion | Generator location, mass, cooling/exhaust, fuel, vibration, load steps. Separate hotel demand from wheel-power requirement | Power + locomotive integrators. Before package selection |
| IC10 Power/coaches | Voltage, phase/frequency, trainline connector ratings, earth continuity, shore changeover. Ratings and topology must be coordinated | Electrical engineer + equipment suppliers. Before cable/breaker selection |
| IC11 Body/HVAC | Equipment cutouts, ducts/returns, fresh air, exhaust separation, drains, service access. Load budget precedes equipment selection | HVAC + body designers. Before body shell fabrication |
| IC12 Passenger circulation | Occupied knees/coats, seat pitch, doorway approaches, handholds, emergency exits. 32-place regular mockup, capacity not approved | Vehicle/accessibility designers + owner. Before seating count freeze |
| IC13 Accessible boarding | Mobility device and user envelope, total load, turning/securement, bridge/lift doorway fit. 42 in side door; large-lift alternative conflicts | Access designer + boarding supplier. Before door frame release |
| IC14 Stairs/swept envelope | Deployed/stowed stairs, guards/gates, locked indication, ground/trackside clearance. 44 in end platform space study | Vehicle + track/station designers. Before stair fabrication |

| ID / interface | Definition and current basis | Owner / gate |
|---|---|---|
| IC15 North Pole station | Same accessible boarding/rescue standard as origin; position and civil works deferred. Passengers disembark; location supplied later | Owner + civil/accessibility designers. Before site-specific construction |
| IC16 Manufacturing/QA | Materials, certificates, WPS/PQR, welder qualifications, NDT, dimensional control, coatings. No member/weld/fastener final sizes released | Responsible engineer + qualified fabricator/inspector. Before structural fabrication |
| IC17 Pilot track/crew | Approved work scope, survey control, rail tools, gauge/crosslevel/twist measurements, independent QA. Farm-crew participation intended; target 0% longitudinal grade, survey verification and drainage required | Track/civil engineer + owner + inspector. Before crew production |
| IC18 Commissioning/release | Static and dynamic test plan, load ballast, instrument calibration, stopping/holding/fault/rescue acceptance. Criteria set by responsible professionals | Integrators + owner + authorities/insurer. Before passengers |

# 10 | Work packages and next release

| Work package | Useful work now | Release needed for permanent work |
|---|---|---|
| Full-size occupied mockup | Tape the actual floor outline and door/seat/route envelopes on level ground. Use stable rated chairs/fixtures for six-adult and mobility-device trials. Record changes to pitch, width and access before altering CAD. | Vehicle/accessibility review of the occupied mockup; do not assume the taped plan establishes evacuation or rated occupancy. |
| Supplier definition | Use the component schedule to obtain certified drawings, exact model/options, loads, duty/temperature ratings, mass/CG, failure behavior, installation requirements and test evidence. | Owner authorization before outreach/orders; engineer acceptance of compatibility and responsibilities. |
| First chassis/coach | Prepare fabrication travelers, material traceability, jig/datum strategy, fit-up sequence, corrosion plan and inspectable weld access. | Issued structural drawings, accepted materials, WPS/PQR, qualifications, inspection plan and critical tolerances. |
| Pilot track | Plan material handling, a demonstrated rail-bending/fastening method, survey control and inspection records. Inventory farm equipment without assuming suitability. | Approved rail/wheel interface, survey/civil design, permitted crew scope, specialist method and independent inspection. |
| Repeated fabrication | Capture first-article dimensions, actual tare/axle loads, assembly time, defects and service access before building repeated cars. | Engineering acceptance of first-article results and controlled revisions. |
| Commissioning | Prepare instrument lists, ballast plan, defect register, no-power recovery and accessible rescue demonstrations. | Engineer-set acceptance criteria; static/dynamic/fault tests, authority and insurer requirements, completed manuals and trained staff. |

The next issue should close selected OEM interfaces and measured mockup results, then develop member sizes, connections, detailed manufacturing drawings, wiring/hydraulic schematics, materials/parts schedules and test limits. Final site documents remain sequenced after the owner supplies the North Pole location and a survey is available.

No supplier has been contacted, no order placed and no project-specific engineering approval obtained by this issue. The authorized supplier site contains the technical release only.

# 11 | Primary source register

Access date: 17 October 2026 unless a record states otherwise. Product publications support only their stated product and conditions. Availability, current quotation, project suitability and signed engineering remain unconfirmed unless explicitly recorded.

## FR01 | Alan Keef - Leisure, Park & Farm; Lappa Valley / Pairi Daiza case studies

Manufacturer describes proprietary passenger bogies, custom narrow-gauge coach design and 2024 weather-protected accessible carriage delivery. Relevant page sections: Lappa Valley and Pairi Daiza.

Limit: No published 914.4 mm bogie capacity, exact interface outline, standalone availability or price for this project. Other gauges and projects do not prove compatibility.

https://www.alankeef.co.uk/services/leisure-park-farm

## FR02 | Severn Lamb - Jupiter manufacturer technical table

Jupiter table: gauge914 mm, minimum radius50 m, maximum gradient1%, maximum speed16 km/h, diesel/LPG, open or enclosed passenger compartments.

Limit: These are vendor whole-train published figures, not ratings for custom cars or standalone bogies. 914 mm published rounding must be reconciled with exact36in=914.4 mm controlled interface. No model-specific axle, frame, CG or suspension ratings found.

https://www.severn-lamb.com/people-movers/tracked-trains/rail-trains/jupiter-2

## FR03 | GHH-BONATRANS - Wheelsets and Design of new products

Manufacturer offers customer-specific wheelset drawings and design verification by calculations and test programs; wheels/bogie/drive/brake interfaces are coordinated. Sections: Introduction of design department; Elaboration of drawings; Wheelset solutions.

Limit: No quoted small-order36in wheelset, component dimensions, loads, availability or price.

https://bonatrans.com/en/research-and-development/design-of-new-products/

## FR04 | Schaeffler FAG - Axlebox Bearings for Passenger Cars & Locomotives

Axlebox Bearing Housing Units section describes application-specific FEM housings, WJ/WJP/cylindrical/TAROL units, factory sealed/greased units, EN12080 manufacture and EN12082 performance testing.

Limit: Product-family capability only; no selected part or C/C0/life rating. Climatic and duty compatibility must be confirmed with vendor.

https://www.schaeffler.us/us/products-and-solutions/industrial/industry_solutions/rail/axlebox_bearings_passenger_cars_locomotives/

## FR05 | Hydro - Alloy6005A data sheet, revision2019/02

Page1 lists rail components as an application, density0.098lb/in3, welding/brazing and galvanic-corrosion caution. Page2 gives temper/thickness-dependent minimum base-material properties.

Limit: Published2019 sheet remains publicly offered; not a project-specific material certificate or welded structural allowable stress. Confirm ordered grade/temper/thickness and current mill certification.

https://www.hydro.com/globalassets/01-products--services/extruded-profiles/americas/ena-resources/alloy-data-sheets/hydro_2019_data_sheet_6005a_revise_1.pdf

## FR06 | Aluminum Association - Aluminum Design Manual2020 and Welding Aluminum

Publisher identifies weld-affected strengths, post-weld heat-treated strengths, structural design provisions and welding-quality guidance.

Limit: Public descriptions only; full manuals not purchased/read. No claim this establishes the adopted project code or all current editions.

https://www.aluminum.org/news/aluminum-association-releases-2020-edition-aluminum-design-manual

## FR07 | NEN-EN12663-1:2010+A2:2023 - Railway vehicle body structural requirements

Official standards-body scope covers body loads, materials and validation by analysis/testing; recognizes vehicles not fitting standard categories need specified requirements. Listed current.

Limit: Public scope only; complete standard not read. European scope is not proof of Nevada applicability. Adopted design criteria require engineer/authority agreement.

https://www.nen.nl/en/nen-en-12663-1-2010-a2-2023-en-318525

## FR08 | NEN-EN13749:2021+A1:2023 - Bogie frame structural requirements

Official scope covers bogie frames, bolsters, axleboxes and functional attachments, including design, assessment, verification and manufacturing quality. Listed current with2026 draft successor indicated.

Limit: Public scope only, no numerical criteria imported; current edition to reconfirm at design freeze. Not automatically governing in Nevada.

https://www.nen.nl/en/nen-en-13749-2021-a1-2023-en-318518

## FR09 | EVS-EN14363:2016+A2:2022 - Running behaviour and stationary tests

Official standards body identifies testing/simulation for acceptance of railway running characteristics and stationary tests.

Limit: Title/history/scope only. No numerical wheel-unloading criteria adopted. Suitability for this unusual low-speed private railway requires engineering decision.

https://www.evs.ee/en/evs-en-14363-2016-a2-2022-consolidated

## BC01 | Rail vehicle actuators / HydroAct

HC1P50 spring actuator; two hydraulic release circuits; up to100kN per disc; max disc width100mm; approx60kg excluding bracket. HSC1P10N listing20kN,45mm disc,15kg.

Limit: Public listing checked only; price, lead time, quotation, current controlled drawing and project suitability unconfirmed

https://rail.knorr-bremse.com/en/us/portfolio/products-and-systems/braking-systems/actuation/

## BC02 | HydroControl

Smart typeu3HU pressure/spring actuated;0-100bar; accumulator version capable of wheel-slide protection. BasicHGK lacks accumulator.

Limit: Public listing checked only; price, lead time, quotation, current controlled drawing and project suitability unconfirmed

https://rail.knorr-bremse.com/en/hk/portfolio/products-and-systems/braking-systems/control/hydraulic-control/

## BC03 | GMXSH DS2610

XSH7.5:11.2kN tangential braking force,65bar minimum full retraction,120bar max pressure,13.62kg caliper/thruster. Conditioned pad nominal friction0.4. Not railway qualification.

Limit: Public listing checked only; price, lead time, quotation, current controlled drawing and project suitability unconfirmed

https://www.twiflex.com/-/media/project/altramotion/shared/files/literature/brand/twiflex-limited/related/product-sheets/gmx-series/ds2610.pdf?rev=939e126f1f99407e92ae8f3d35e7ab0b

## BC04 | GMX series

Minimum disc diameter610mm; actual installation not established.

Limit: Public listing checked only; price, lead time, quotation, current controlled drawing and project suitability unconfirmed

https://www.twiflex.com/products/disc-braking-systems/direct-acting-calipers/gmx-series

## BC05 | People Mover semi-permanent coupler

Double-acting damper,pivot anchor,cable support; no force or dimension on page.

Limit: Public listing checked only; price, lead time, quotation, current controlled drawing and project suitability unconfirmed

https://www.dellner.com/products/semi-permanent-couplers/semi-permanent-coupler-people-mover

## BC06 | 2022 brochure,PDFp19 printed38-39

Semi-permanent modular architecture includes socket joints,draft gear,pivot anchor and central gas-hydraulic buffer.

Limit: Public listing checked only; price, lead time, quotation, current controlled drawing and project suitability unconfirmed

https://dellner.com/wp-content/uploads/2022/08/DellnerBrochure2022_ENG_v1-1.pdf

## BC07 | Railway couplers and connections

Type430/530 compact railway couplers advertised up to300kN compression/tension. Semi-permanent types can incorporate energy absorption. No project compatibility established.

Limit: Public listing checked only; price, lead time, quotation, current controlled drawing and project suitability unconfirmed

https://www.voith.com/corp-en/connection-components-couplings/railway-couplers.html

## BC08 | Emergency Brake Valve

Vents brake pipe for emergency quick action; linked26C controls brake-pipe pressure for car/locomotive application/release.

Limit: Public listing checked only; price, lead time, quotation, current controlled drawing and project suitability unconfirmed

https://www.wabteccorp.com/locomotive/braking-systems/pneumatic-controls/emergency-brake-valve

## BC09 | Braking calculations

Distinguishes holding,dynamic and continuous-duty requirements; considers delay,external load,temperature,power dissipation and pad life.

Limit: Public listing checked only; price, lead time, quotation, current controlled drawing and project suitability unconfirmed

https://www.twiflex.com/braking-calculations

## PU01 | Severn Lamb JUPITER catalog

{"gauge_mm": 914, "capacity_people": [160, 320], "maximum_gradient_percent": 1, "minimum_radius_m": 50, "maximum_speed_kmh": 16, "drive_options": ["diesel", "LPG"], "compartments": ["open", "enclosed"]}

Limit: Catalog configuration, not custom train rating. No mass, power or TE curve published.

https://www.severn-lamb.com/people-movers/tracked-trains/rail-trains/jupiter-2

## PU02 | Western Train locomotive catalog - separate Jupiter

{"drive_wheel_diameter_in": 38, "power_location": "tender", "brakes": "air", "photo_label_gauge_in": 24}

Limit: Photo gauge is not proof of a 36-inch variant or proposed-train performance.

https://www.westerntrain.com/locomotives

## PU03 | Western Train FAQ

{"diesel_propulsion": "diesel engines paired with Eaton hydraulic pumps", "gauge_offerings_in": [15, 24, 36], "passenger_brakes": "air"}

Limit: No exact engine, pump, TE, mass or duty ratings; current written specification required.

https://www.westerntrain.com/faq

## PU04 | Cummins F3.8 Tier 4 Final/Stage V North America

{"top_listed_power_kW": 115, "peak_torque_Nm": 620, "peak_torque_rpm": 1600}

Limit: Off-highway product listing does not establish locomotive application/emissions approval; current calibration, net installation rating and derating required.

https://www.cummins.com/en-na/engines/products/f38-tier-4-final-stage-v

## PU05 | Cummins rail product guide - F3.8 intermodal and track-maintenance ratings

{"F3_8_154_kW": 115, "rated_rpm_options": [2200, 2500], "B4_5_200_kW": 149, "B4_5_200_rpm_options": [2200, 2500]}

Limit: Historical manufacturer guide; not a current quotation or application approval. Do not adopt older 129-kW F3.8 rating over current North American page.

https://mart.cummins.com/imagelibrary/data/assetfiles/0059591.pdf

## PU06 | Danfoss H1P 060/068 data sheet AI152886482421en-000901

{"H1P060_displacement_cc_rev": 60.4, "rated_speed_rpm": 3500, "max_working_bar": 420, "maximum_bar": 450, "dry_kg_excluding_PTO_filter": 50}

Limit: Maximum working is not an assigned continuous operating point; pressure above working requires OEM approval. Exact control code and pump maps unresolved.

https://assets.danfoss.com/documents/latest/498340/AI152886482421en-000901.pdf

## PU07 | Danfoss H1B technical information BC152886483576en-001303

{"H1B080_max_displacement_cc_rev": 80, "min_displacement_cc_rev": 16, "theoretical_Nm_per_bar": 1.27, "rated_rpm_at_max_displacement": 3200, "max_working_bar": 450, "SAE_dry_mass_kg": 34.8}

Limit: Corner power not continuous system rating. Fluid case fill, pressure/speed, case drain and independent brakes are required; rail application unresolved.

https://assets.danfoss.com/documents/latest/534145/BC152886483576en-001303.pdf

## PU08 | Allison 1000/2000 series product family

{"family_power_kW": [154, 272], "family_max_torque_Nm": 950}

Limit: Family maxima cannot select one model or approve rail reverse/stall duty. Exact model remains HOLD.

https://www.allisontransmission.com/applications--products/products/1000-series-2000-series

## PU09 | Danfoss EM-PMI240-T180 data sheet (returned revision AI355256833496en-000208)

{"selected_study_variant": "EM-PMI240-T180-4400", "class_F_at65C_continuous_kW": 80, "class_F_at65C_continuous_Nm": 175, "nominal_rpm": 4400, "max_rpm": 8800, "nominal_line_line_VAC": 500, "minimum_coolant_L_min": 20, "nominal_coolant_C": 65, "max_coolant_C": 70, "derating_above_C": 65}

Limit: URL suffix is older but served PDF states July 2026 revision 000208. Converter, DC link, generator and railway duty not selected. Thermal class and coolant values must remain attached to rating.

https://assets.danfoss.com/documents/latest/456958/AI355256833496en-000205.pdf

## PU10 | Danfoss EC-C1200-450 user guide (returned revision BC265735231757en-000211)

{"study_variant": "EC-C1200-450+MC180", "nominal_current_ARMS": 180, "nominal_power_kVA": 150, "nominal_DC_link_V": 750, "DC_link_range_V": [0, 850], "coolant_max_C": 65, "coolant_min_flow_L_min": 10, "MC350_coolant_max_C": 40}

Limit: URL suffix 000209 served PDF revision 000211 dated 2026. Retrieved directly after web size limit. MC350 has 40C coolant limit and is not substituted for MC180. Bus/motor/cooling/protection/control matching requires supplier confirmation.

https://assets.danfoss.com/documents/latest/591751/BC265735231757en-000209.pdf

## PU11 | Gmeinder narrow-gauge axle gearboxes GGT190EHA/497 +190E/327

{"reference_gauge_mm": 760, "max_input_torque_Nm": 12150, "max_input_rpm": 2720, "ratios": [4.97, 3.27]}

Limit: Reference railway gearbox only, not 914-mm compatible selected wheelset or axle assembly.

https://www.gmeinder.com/en/gearboxes/gearboxes-for-locomotives/narrow-gauge-locomotive-axle-gearboxes-ggt190eha/497-190e/327-12983

## PU12 | Konvekta rail HVAC overview - HVAC 5401 page 23

{"cooling_kW": 16, "rating_condition": "ta=ti=40C", "heating_kW": 7.2, "supply_VAC": 400, "control_VDC": 24, "mass_kg": 275, "optional_mass_kg": 240, "dimensions_mm": [1500, 1500, 370]}

Limit: Not high-ambient 110F/75F capacity. Current US configuration/refrigerant and 60Hz compatibility unresolved.

https://www.konvekta.de/fileadmin/user_upload/docs/datenblaetter/schiene/Produktuebersicht_Schiene_dt-engl_eBook_WEB_0821.pdf

## PU13 | Northwest Rail Electric NW-64-CN condenser

{"nominal_cooling_tons_approximately": 5, "application": "rail, narrow-gauge and tourist cars"}

Limit: Condenser only; complete matched high-ambient/latent performance, evaporator, ventilation, heating, power, size and mass must be quoted.

https://www.nwrail.com/nwre_products/64cn.html

## PU14 | Stadco Mark 6 sales specifications S32.25 Rev 01/25

{"frequency_Hz": 60, "three_phase_voltage_option": [240, 480], "6-42T4F19": {"intermittent_kW": 42, "prime_kW": 40, "continuous_kW": 38, "dimensions_in": [72, 32, 30], "approx_mass_lb": 1790}, "6-80T4F19": {"intermittent_kW": 80, "prime_kW": 74, "continuous_kW": 70, "dimensions_in": [103, 36, 32], "approx_mass_lb": 3126}, "reference_ambient_F": 80, "reference_altitude_ft": 500}

Limit: Vendor climate derating required; 208V values differ. Motor-start kVA quoted at 35% dip is not proof that loads tolerate dip. Cylinder-count inconsistency not repeated. No current quote/lead time.

https://stadcogen.com/wp-content/uploads/2025/05/S32.25-Mark-6-Specs.pdf

## PU15 | Stadco Railgen current product page

{"listed_models": ["6-42T4F", "6-80T4F"], "6-50T4F": "proposed / under development"}

Limit: Model 50 not treated as available. Current manufacture/lead time not confirmed.

https://stadcogen.com/diesel-generators/under-car-generators-for-rail-cars/

## PU16 | Cummins T-030 liquid-cooled generator application manual

{"guidance_topics": ["load sizing", "motor starting", "duty rating", "site conditions", "protection and installation"]}

Limit: General engineering guidance, not approval of proposed rail electrical system.

https://www.cummins.com/sites/default/files/2024-08/t030-Liquid-cooled-gen-set-application-manual.pdf

## PU17 | Severn Lamb Dayton project completion

{"gauge_ft": 3, "traction": "battery electric", "motor_count": 4, "motor_kW_each": 30, "reported_design_train_lb": 150000, "reported_design_speed_mph": 6, "reported_design_grade_percent": 3}

Limit: Custom locomotive; cannot upgrade diesel Jupiter specification. Source battery unit 220kW ambiguous and not converted to kWh.

https://www.severn-lamb.com/dayton-celebration

## BB01 | US Access Board - ADA Chapter 10, sections 1002.2-1002.4

Amusement-ride rules specify approach, wheelchair space, companions, direct platform gaps and alternative bridge/ramp references. Side-entry advisory gives 42 in opening with 60 x 36 in maneuvering example.

Limit: Applicability to this train is unresolved; thresholds are not a blanket vehicle approval. Do not substitute advisory geometry for actual maneuvering verification.

https://www.access-board.gov/ada/chapter/ch10/

## BB02 | US Access Board - Chapter 10 Amusement Rides guide

Guide explains accessible load/unload approach and ramps/bridge plates under the referenced transportation-vehicle provisions.

Limit: Confirm legal classification, scoping, any exemptions and complete train/site duties with authorities; guide alone is not the determination.

https://www.access-board.gov/ada/guides/chapter-10-amusement-rides/

## BB03 | Coosa - Bluewater Panels Imperial Technical Data Sheet

Bluewater26 nominal26lb/ft3; 4x8x3/4 in panel approximate52lb; fiberglass-reinforced polyurethane; standard panel FMVSS302 claim; material data typical, not application guarantee.

Limit: No approved train floor thickness/span, fastener strength allowable or complete fire assembly. Standard and fire-rated product variants must remain distinct.

https://coosacomposites.com/wp-content/uploads/2024/10/CoosaComposites_BluewaterPanels-Imperial.pdf

## BB04 | Coosa - Flame Spread and Smoke Rated Panels

Separately ordered fire-rated Bluewater and Nautical lines; ASTM E84-23 test dated2024-09-23; published Bluewater flame spread10/smoke300 and ClassA claim.

Limit: E84 is not a rail or amusement assembly approval or proof of smoke toxicity compliance; obtain exact variant certificates.

https://coosacomposites.com/flame-spread-rated-panels/

## BB05 | Plascore - Base Panels technical table

AA3.6-80 aluminum-skin/aluminum-core typical0.84lb/ft2 at0.50in and0.94lb/ft2 at0.75in. Open edges; manufacturer offers custom panels.

Limit: Typical test data; not structural allowables. Edge closure, inserts, skin support, adhesive/fire system and concentrated loads require project engineering.

https://plascore.com/composite-panels/base-panels/

## BB06 | Armacell - ArmaFlex Rail SD technical specifications

Declared conductivity <=0.040W/(m K) at0C; vapor diffusion factor>=5000; published ASTM E162/E662 rail test claim limited to3-13mm; UV protection required; product-specific certified adhesives listed.

Limit: No weight/density adopted. Do not extend thin-product tests to thicker layered wall assembly; regional availability and complete US fire basis unconfirmed.

https://www.armacell.com/en-NL/product/technical-specs/armaflex-rail-sd

## BB07 | Sika - Sikaflex-252 US Product Data Sheet, Version04.01 April2025

Vehicle assembly adhesive; actual-substrate adhesion/compatibility tests required; application10-35C; typical tensile lap shear2.5MPa; moisture cured.

Limit: Typical shear value is not allowable design stress. Surface prep, geometry, peel/fatigue/environment, cure and fire acceptance need qualified process.

https://industry.sika.com/dam/dms/us01/o/sikaflex-252.pdf

## BB08 | Gerflor - Tarabus Sirius technical data sheet

NT2.25+/-0.15mm and2.20+/-0.15kg/m2; MK6.20+/-0.30mm and3.65+/-0.30kg/m2; R10 slip rating; FMVSS302 and ASTM E648Class1 statements.

Limit: Bus flooring candidate; assembly, substrate adhesive, wet farm dirt performance and applicable rail/fire requirements unapproved.

https://cdn.gerflor.com/media/2/38529/tarabus%20sirius%20-%20technical%20datasheet.pdf

## BB09 | BraunAbility - NCL1000 Century Series Application Guide, cover 6-3-26 (3 June 2026)

Model NCL1000IB3754HB-2 row:385lb mass,1000lb rating,37x54in platform,66.06in overallheight,48intravel,50.96in overallwidth,45.85in minclear doorwaywidth,67in minclear height,18in thresholddepth. 12VDC/max70A; manual occupied raise/lower and fold/unfold backup.

Limit: Document cover6-3-26 is interpreted as3June2026 (US manufacturer); all dimensions reference only. Motor-vehicle lift certification does not approve rail mounting. Current42in side door conflicts.

https://www.braunability.com/content/dam/braunability/docs/comm/na/app-guide/updated-dpa-lift-app-guides/Lifts-Comm-Century-1000-Guide-6-3-26.pdf

## BB10 | BraunAbility - NCL1000-2 brochure,405245

Manufacturer explicitly cautions that1000lb side-mounted lift load can cause body roll and requires suspension/platform-tilt assessment.

Limit: Older2019 brochure supplement; use2026 model-specific application table for dimensions and obtain latest manual at order.

https://www.braunability.com/content/dam/braunability/docs/comm/na/bro/405245-NCL1000-2-Lift-Flyer.pdf

## BB11 | Ricon/Wabtec - Rail-car PowerLift

Specifically marketed rail-car lift with660lb/300kg published load rating.

Limit: Dimensions, mass, travel, manual backup and compatibility unverified.295kg stress rider/device leaves5kg nominal capacity headroom; not selected.

https://www.riconcorp.com/pages/products_rail_car_powerlift.asp

## BB12 | Q’Straint - QRT-360 wheelchair and occupant securement

Four-point retractable system; rail applications listed; manufacturer WC18/WC19 and30mph/20G test claims; Q-10007 kit includes4 L-track retractors and retractable lap/shoulder combination.

Limit: Must obtain complete anchorage/installation, actual wheelchair/rider mass limits, orientation and compatibility. Catalog test is not approval of custom coach anchor backing.

https://www.qstraint.com/qrt-360/

## TD01 | Clark County Ordinance 5378 - 2026 Amusement Ride and People Mover Code

Current County code; installation-duration scope, Building Official classification, design responsibility, QA and commissioning provisions read.

Limit: No project classification or County interpretation received. Standards applicability/adopted edition matrix unresolved.

https://www.clarkcountynv.gov/adobe/assets/urn:aaid:aem:c5347a68-d8cb-444a-9cf2-e9bf7b12080f/original/as/ORD-2026-05-19-Clark-County-Ordinances-5378.pdf

## TD02 | 49 CFR Part 209 Appendix A - FRA safety-jurisdiction policy

Tourist operation policy includes less-than-24-inch and off-system/insular exceptions; four stated non-insular features reviewed.

Limit: No parcel/crossing/proximity plan or project-specific applicability position available.

https://www.ecfr.gov/current/title-49/subtitle-B/chapter-II/part-209/appendix-Appendix%20A%20to%20Part%20209

## TD03 | Nevada Revised Statutes Chapter 455B - Recreational Parks

Amusement park/ride definitions and operator safety duty reviewed.

Limit: Does not itself provide a rail design approval or resolve the whole jurisdictional allocation.

https://www.leg.state.nv.us/nrs/NRS-455B.html

## TD04 | ASTM F2960-23 - Permanent Amusement Railway Ride Tracks and Related Devices

Public scope covers permanent amusement track >=12-inch gauge, excludes rolling stock, identifies subsystem interface coordination and FRA/portable exclusions.

Limit: Public scope/abstract only; full licensed standard not obtained or read. Not claimed as County-adopted.

https://store.astm.org/f2960-23.html

## TD05 | ASTM F3054-26 - Operations of Amusement Railway Rides, Devices, and Facilities

Active page updated March 16, 2026; operations scope >=12-inch gauge with stated portable and FRA exclusions.

Limit: Public scope only; full licensed standard not obtained or read. Adopted edition unresolved.

https://store.astm.org/f3054-26.html

## TD06 | Clark County current building codes and amendments

Lists current 2026 ARPM code, January 11, 2026 adoption of 2024 building codes, 2023 NEC and local amendments.

Limit: Listing is not a project-specific facility or vehicle code determination.

https://www.clarkcountynv.gov/government/departments/building___fire_prevention/codes/

## TD07 | 49 CFR 37.37 - Other applications

Distinguishes private transportation/shuttle services, primarily recreational conveyances, and private rail passenger-car provisions.

Limit: North Pole disembarkation is relevant service information, not by itself the legal classification.

https://www.ecfr.gov/current/title-49/subtitle-A/part-37/subpart-B/section-37.37

## TD08 | Nevada Administrative Code 625 - Professional Engineers and Land Surveyors

Responsible-charge, document status, discipline and signature/seal requirements in 625.610-.613 reviewed.

Limit: No licensed professional has adopted or approved this preliminary package.

https://www.leg.state.nv.us/nac/NAC-625.html

## TD09 | U.S. Access Board ADA Standards Chapter 10 - Recreation Facilities

Section 1002 addresses amusement-ride boarding and wheelchair-space interfaces.

Limit: Applicability to this operation remains open; not used as substitute for determining transportation requirements.

https://www.access-board.gov/ada/chapter/ch10/

## TD10 | JENNMAR Rail Benders

Lists #3H-25 hydraulic rail bender, 25-ton, 25-60-pound rail range.

Limit: Page general descriptions are inconsistent. Exact profiles, units, grade, instructions, application acceptance, stock and lead time unverified. Candidate only.

https://www.jennmar.com/products/rail-benders

## TD11 | Clark County Fire Prevention

Division identifies plan review/inspection scope including fire access, fuel storage/use and repair garages.

Limit: Parcel-specific lead and response arrangements require confirmation.

https://www.clarkcountynv.gov/government/departments/fire/fire-prevention

## TD12 | Clark County Public Works

Development Review identifies drainage, technical, traffic and land-use review, encroachment permitting and off-site construction inspection.

Limit: Does not resolve whether each proposed on-farm element requires that approval.

https://www.clarkcountynv.gov/government/departments/public_works_department/

## TD13 | Clark County zoning and use permit guidance

Identifies Comprehensive Planning as contact for zoning/use permit questions and describes use/condition modification pathways.

Limit: Existing farm SUP and conditions were not reviewed in this subtask; no entitlement claimed.

https://www.clarkcountynv.gov/government/departments/administrative_services/code_enforcement_-_public_response_office/zoning-violations

## TD14 | Moapa Valley Fire District - County directory page

County maintains a specific Moapa Valley Fire District page.

Limit: Does not establish parcel-specific plan-review versus emergency-response roles; coordinate with County Fire Prevention.

https://www.clarkcountynv.gov/government/departments/fire/moapa-valley-fire-district-advisory-board